



قسم علوم
الحاسبات

Nassin (نصّين): Bridging the Gap to Formal Arabic

Real-Time Dialect-to-MSA Translation using AraT5



عربثون
ARABTHON

Modern Standard and Classical Arabic Track

ABSTRACT

The Arabic language is characterized by a significant linguistic gap between "Ammiya" (spoken dialects) and "Fusha" (Modern Standard Arabic - MSA). While formal Arabic is strictly required for professional, legal, and educational communications, most daily interaction occurs in informal dialects. This creates a productivity barrier, as users often struggle to instantly convert their natural thoughts into formal text.

To address this, we developed a specialized solution by fine-tuning the AraT5 Transformer architecture. Unlike general-purpose Large Language Models (LLMs) which often misinterpret regional slang, our model was trained on a curated dataset of 26,000 dialect-MSA pairs. The system acts as a real-time "Formalizer," correcting grammar and replacing colloquialisms with proper terminology while preserving the original intent. This project demonstrates the efficacy of specialized models in bridging the gap between low-resource dialects and formal linguistic standards.

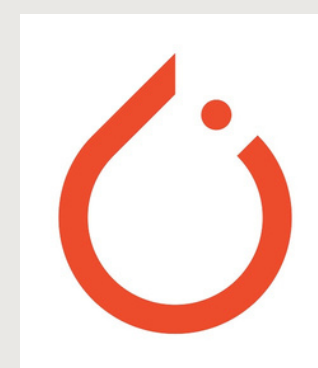
IDEA FIELDS

This project sits at the intersection of Natural Language Processing (NLP) and Business Productivity.

- **Corporate Sector:** It serves as an essential tool for drafting formal emails, reports, and meeting minutes, saving time for employees who are more comfortable speaking in dialect.
- **Customer Service:** It automates the processing of customer complaints written in slang, converting them into structured formal data for analysis.
- **Education:** It acts as a learning aid for students, helping them transition from colloquial speech to academic writing standards.
- **Government Services:** It facilitates clearer communication between citizens (who speak dialect) and official digital portals (which require standard Arabic).

DATASET DETAILS

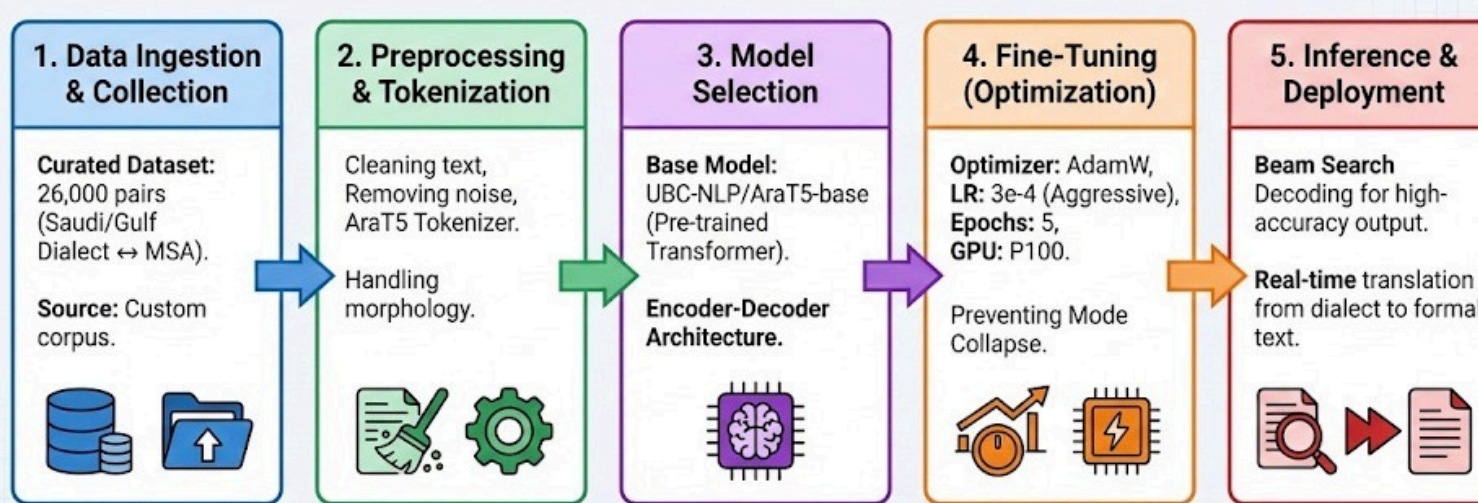
- A dialect-to-MSA dataset was collected from Composite Corpus (PADIC + NADI) and processed into a structured format.
- The data was cleaned, sampled, and split into training and validation sets.
- Size: around 26,000 sentences pairs
- A pre-trained T5 transformer model was fine-tuned for the task of dialect-to-MSA translation.
- The model was trained for multiple epochs and evaluated using sample predictions.



FUTURE WORK

- **Voice Integration (ASR):** Adding speech-to-text so users can speak in dialect and receive formal text outputs.
- **Browser Plugin:** Deploying as a Chrome extension for real-time email correction in Gmail/Outlook.
- **Multi-Dialect Support:** Expanding the dataset to cover Egyptian and Levantine dialects.

Methodology: Dialect-to-Formal Arabic Translation Pipeline



A specialized pipeline optimized for low-resource Arabic dialect transformation.

RESULTS

- **Validation Loss:** Reduced from 0.60 to 0.27.
- **Optimization:** Solved "Mode Collapse" using 3e-4 Learning Rate.
- **Accuracy:** Successfully resolves Saudi negation (e.g., "لا يستجيب" → "مو راضي").
- **Inference:** Real-time translation on P100 GPU.

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